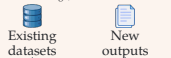


a

Disagreement-based preference mining

Open-ended tasks



Metrics and judges

Signal 1 Signal 2 Priority
 $A > B = A > B$ Lower

$A > B \neq B > A$ Higher

Conflict → rank higher
 + random sample



Reuse labels
or ask humans
for new ones

Human feedback

Signals can share errors

b

Depth-first evaluator evolution

Test predictions; adapt edit scope; retain best



Active direction: make layout errors count

Contract: detect faults; keep scores stable

Base model fixed

What should we test next?

Candidate C



Add a layout
detector

Working-state transition
 Predicted change; no loss
 Else: keep parent



Retain diagnostics

Observed

Layout errors detected.
Final scores unchanged.

Question to test

Do findings reach scoring?

Direction continuation

Revise working W



Pass findings
into scoring

Net / declared repairs
Adapt edit budget Δ

Stop reason → new direction



Promotion feedback only

Working program W

Periodic checkpoint of W

Global checkpoint selection



Full Val

Val gain $\geq$ margin
→ update best program

Otherwise → keep best program

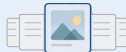


Best program

c

Downstream applications

Output selection



Rank and select

Critic-guided refinement



Reward-guided training



Reward
Planned

Best program fixed after search

Constructed example, not an observed trajectory